THE POLYNOMIAL PIECE OF THE MOR SPACE IS SEPARATED OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1898-1911 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1913-1962, the complete author proof (48 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_moduli.json (pb=2158, stmt 1898-1911, proof 1913-1962).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DPN: sources/stacks-moduli/pinned-original-source.tex, source lines 1843–1846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPI: sources/stacks-moduli/pinned-original-source.tex, source lines 1476–1486. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B8W: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZQ: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081F: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MT: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNK: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPQ: sources/stacks-moduli/pinned-original-source.tex, source lines 1870–1896. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D31: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KG: sources/stacks-moduli/pinned-original-source.tex, source lines 1962–1962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *With B, X, Y as in the introduction of this section, let $\mathcal{L}$ be ample on X/B and let $\mathcal{N}$ be ample on Y/B. See Divisors on Spaces, Definition 0D31. Let P be a numerical polynomial. Then*

$$Mor_B^{P,\mathcal{M}}(Y, X) \longrightarrow B$$

is separated and of finite presentation where $\mathcal{M} = pr_1^*\mathcal{N} \otimes_{\mathcal{O}_{Y\times_B X}} pr_2^*\mathcal{L}$.

Proof. By Lemma 0DPN the morphism $Mor_B(Y, X) \to B$ is separated and locally of finite presentation. Thus it suffices to show that the open and closed subspace $Mor_B^{P,\mathcal{M}}(Y, X)$ of Remark 0DPQ is quasi-compact over B.

The question is étale local on B (Morphisms of Spaces, Lemma 03KG). Thus we may assume B is affine.

Assume $B = \mathrm{Spec}(\Lambda)$. Note that X and Y are schemes and that $\mathcal{L}$ and $\mathcal{N}$ are ample invertible sheaves on X and Y (this follows immediately from the definitions). Write $\Lambda = \mathrm{colim}\, \Lambda_i$ as the colimit of its finite type $\mathbf{Z}$-subalgebras. Then we can find an i and a system $X_i, Y_i, \mathcal{L}_i, \mathcal{N}_i$ as in the lemma over $B_i = \mathrm{Spec}(\Lambda_i)$ whose base change to B gives $X, Y, \mathcal{L}, \mathcal{N}$. This follows from Limits, Lemmas 01ZM (to find X_i, Y_i), 0B8W (to find $\mathcal{L}_i$, $\mathcal{N}_i$), 01ZQ (to make $X_i \to B_i$ separated), 081F (to make $Y_i \to B_i$ proper), and 09MT (to make $\mathcal{L}_i$, $\mathcal{N}_i$ ample). Because

$$Mor_B(Y, X) = B \times_{B_i} Mor_{B_i}(Y_i, X_i)$$

and similarly for $Mor_B^P(Y, X)$ we reduce to the case discussed in the next paragraph.

Assume B is a Noetherian affine scheme. By Properties, Lemma 0DNK we see that $\mathcal{M}$ is ample. By Lemma 0DPI we see that $\mathrm{Hilb}^{P,\mathcal{M}}_{Y\times_B X/B}$ is of finite presentation over B and hence Noetherian. By construction

$$Mor^{P,\mathcal{M}}_B(Y,X) = Mor_B(Y,X) \cap \mathrm{Hilb}^{P,\mathcal{M}}_{Y\times_B X/B}$$

is an open subspace of $\mathrm{Hilb}^{P,\mathcal{M}}_{Y\times_B X/B}$ and hence quasi-compact (as an open of a Noetherian algebraic space is quasi-compact). □